

Machine Learning Projects and Applications

Abstract

This mini-paper presents three machine learning projects on real-world datasets, covering supervised and unsupervised methods such as logistic regression, decision trees, random forests, SVM, PCA, and clustering. The projects span educational outcomes, Parkinson's disease, and social network advertising, focusing on data preprocessing, visualization, and model evaluation. Participation in the EEML Summer School will deepen my understanding of advanced ML methods and provide an opportunity to collaborate with international peers.

Introduction

I am an undergraduate student in Economics at UNEC with a strong interest in Data Science and Machine Learning. I apply ML to real datasets to gain insights and strengthen my skills in preprocessing, EDA, predictive modeling, and model evaluation in Python.

Methods / Projects

1. Education Predictors Dataset

The objective of this project was to understand how different educational features influence student outcomes. I applied dimensionality reduction using PCA, visualized the dataset with a listed colormap, and performed a train-test split for evaluation. Logistic regression was used as a baseline model, achieving an accuracy of 0.66. This project improved my skills in linking preprocessing techniques to predictive performance, understanding the effect of dimensionality reduction on features, and exploring the importance of visualization in data analysis.

2. Parkinson's Disease Dataset

The goal was to explore how voice measurement features relate to Parkinson's diagnosis. Data cleaning was performed to remove duplicates and inconsistencies, followed by extensive exploratory data analysis using KDE plots, histograms, boxplots, violin plots, hexbin plots, countplots, probability plots, and regression plots. Outlier detection and feature scaling were also applied. This project strengthened my understanding of data visualization, feature behavior analysis, scaling importance before modeling, and statistical distribution analysis.

3. Social Network Ads Dataset

This project aimed to predict whether a user would purchase a product based on demographic features. Data preprocessing and train-test split were performed, and a decision tree classifier was applied. The model achieved an accuracy of 0.86, with the confusion matrix highlighting correct identification of actual buyers. Through this project, I enhanced my understanding of classification metrics, model interpretability, feature importance evaluation, and controlling overfitting and model complexity.

The projects applied methods including logistic regression, decision trees, random forests, SVM, PCA, and clustering. The methodology follows best practices described in Géron (2019), Cormen et al. (2009), and Bruce et al. (2020).

Results

- Logistic regression on the Education dataset achieved an **accuracy of 0.66**, showing baseline predictive power.
- PCA successfully reduced feature dimensionality while retaining essential variance in the Education dataset.
- EDA on Parkinson's dataset revealed key patterns in feature distributions and helped identify important preprocessing steps.
- Decision tree classifier on Social Network Ads dataset achieved **accuracy of 0.86**, with the confusion matrix highlighting correct identification of actual buyers.
- Across all projects, visualization and preprocessing were essential for model performance and interpretability.

Conclusion / Motivation

These projects demonstrate practical experience in supervised and unsupervised learning, data preprocessing, and model evaluation. Attending the EEML Summer School will allow me to deepen my knowledge in modern machine learning techniques, gain hands-on experience with advanced methods, and collaborate with motivated international peers. This experience aligns with my long-term goal of pursuing graduate studies in Data Science and building a career focused on solving complex, real-world problems through data-driven insights.

References

1. Géron, A. *Hands-On Machine Learning with Scikit-Learn and PyTorch*. O'Reilly, 2019. (ISBN: 9798341607972)
2. Cormen, T., Leiserson, C., Rivest, R., Stein, C. *Introduction to Algorithms*. MIT Press, 2009.
3. Bruce, P., Bruce, A., Gedeck, P. *Practical Statistics for Data Scientists: 50+ Essential Concepts Using R and Python*. O'Reilly, 2020.